

Multidimensional continuous factorization at the stable-rank scale

Autonomous solution packet for arXiv:1909.00807

11 August 2026

Abstract

Dai–Hore–Jiao–Lan–Motakis ask whether their continuous factorization theorem for an interval extends to a matrix function on $\mathbb{R}^d$, and what dimension can be factored. We give an affirmative answer. If $A : \mathbb{R}^d \rightarrow M_N(\mathbb{R})$ is continuous, $\|A(x)\| \leq 1$, and every column has norm at least θ , then one can continuously factor I_n through $A(x)$, with norm product at most $2/\theta$, for every

$$n \leq \max \left\{ 1, \left\lceil \frac{3N\theta^2}{4 - \theta^2} \right\rceil - d \right\}.$$

The proof combines a trace/singular-value count with a finite-dimensional selection lemma. The d -loss is the obstruction-theoretic cost of choosing a frame over a d -dimensional base.

1 The source question and the result

The source proves the interval case with a smaller bound. Its second question asks explicitly for a version on $\mathbb{R}^d$:

Question 2. For the theorem in the continuous case, we considered $A : I \rightarrow M_N(\mathbb{R})$ where I is an interval of $\mathbb{R}$. We conjecture that a version of Theorem 3.9 is also true for a continuous matrix function $A : \mathbb{R}^d \rightarrow M_N(\mathbb{R})$. What is the relation between d , N , θ , and the dimension n in the conclusion of such a theorem?

We use the source’s stronger column formulation. The diagonal formulation follows because $|a_{ii}(x)| \leq \|A(x)e_i\|$.

Theorem 1 (multidimensional factorization). *Let $d, N \in \mathbb{N}$, let $A : \mathbb{R}^d \rightarrow M_N(\mathbb{R})$ be continuous, and assume*

$$\|A(x)\| \leq 1, \quad \min_{1 \leq i \leq N} \|A(x)e_i\| \geq \theta > 0 \quad (x \in \mathbb{R}^d).$$

Put

$$q = \left\lceil \frac{3N\theta^2}{4 - \theta^2} \right\rceil, \quad m = \max\{1, q - d\}.$$

For every $1 \leq n \leq m$ there are continuous functions

$$R : \mathbb{R}^d \rightarrow M_{N \times n}(\mathbb{R}), \quad L : \mathbb{R}^d \rightarrow M_{n \times N}(\mathbb{R})$$

such that, for every $x \in \mathbb{R}^d$,

$$L(x)A(x)R(x) = I_n, \quad \|L(x)\| \|R(x)\| \leq \frac{2}{\theta}.$$

Thus the answer is linear at the stable-rank scale: for fixed d and $N\theta^2 \rightarrow \infty$, one may take $n \geq (3/4 + o(1))N\theta^2 - d$. No regularity or behavior at infinity is imposed on A .

2 The frame-selection lemma

We record the only topological input. Here $\dim X$ is covering dimension and $\text{St}(n, N) = \{Q \in M_{N \times n}(\mathbb{R}) : Q^*Q = I_n\}$.

Lemma 2 (frames in a lower-semicontinuous field). *Let X be a paracompact metric space with $\dim X \leq d$. Suppose that $x \mapsto E_x$ is a lower-semicontinuous field of linear subspaces of $\mathbb{R}^N$, in the sense that*

$$\{x : E_x \cap U \neq \emptyset\}$$

is open for every open $U \subset \mathbb{R}^N$. If $\dim E_x \geq q$ for all x , then, whenever $1 \leq n \leq q - d$, there are continuous orthonormal vectors $u_1(x), \dots, u_n(x) \in E_x$.

Proof. We use the global finite-dimensional corollary of Michael's selection theorem [1, Theorem 1.2]. In the form needed here it says that if $\dim X \leq d$ and $x \mapsto F_x$ is a lower-semicontinuous field of linear subspaces with $\dim F_x \geq d + 1$, then the unit-sphere carrier $x \mapsto S(F_x)$ has a continuous selection.

For clarity, all of its hypotheses are transparent in this special case. The values $S(F_x)$ are closed subsets of the complete metric space S^{N-1} and are $(d - 1)$ -connected, because they are spheres of dimension at least d . The family of unit spheres of Euclidean subspaces is equi- LC^{d-1} : sufficiently close points in any one of these spheres can be joined and contracted by normalized straight-line homotopies inside a uniformly slightly larger ball. Finally the sphere carrier is lower semicontinuous. Indeed, a nearby nonzero vector in F_x can be normalized, and normalization is continuous away from zero. These are exactly the hypotheses in Michael's finite-dimensional theorem. Paracompactness, not compactness, is the base assumption, so the conclusion applies globally to the noncompact spaces used below.

Apply this result first to E_x . Since $q \geq n + d \geq d + 1$, it gives a continuous unit vector $u_1(x) \in E_x$. Inductively, after choosing $u_1, \dots, u_j$, put

$$E_x^{(j)} = E_x \cap \text{span}\{u_1(x), \dots, u_j(x)\}^\perp.$$

This is again a lower-semicontinuous field: locally choose a vector of E_x near a prescribed vector of $E_{x_0}^{(j)}$ and subtract its continuous orthogonal projection onto the already selected frame. Moreover $\dim E_x^{(j)} \geq q - j$. For $j < n$ this is at least $q - n + 1 \geq d + 1$, so Michael's theorem selects u_{j+1} . The induction ends with the asserted orthonormal n -frame. $\square$

Lemma 3 (spectral-frame selection). *Let X and d be as above, and let $B : X \rightarrow M_N(\mathbb{R})$ be continuous and self-adjoint. Fix $a \in \mathbb{R}$. Suppose that $B(x)$ has at least q eigenvalues strictly larger than a at every x . If $1 \leq n \leq q - d$, there is a continuous $Q : X \rightarrow \text{St}(n, N)$ such that*

$$Q(x)^* B(x) Q(x) > a I_n \quad (x \in X).$$

Proof. Let E_x be the sum of the eigenspaces of $B(x)$ with eigenvalue $> a$. Then $\dim E_x \geq q$. This spectral field is lower semicontinuous even though its dimension may jump. To see this at x_0 , take $v \in E_{x_0}$. Only finitely many eigenvalues occur in its spectral decomposition, so for some $\eta > 0$ it

belongs to the spectral subspace for $(a + \eta, \infty)$. Choose a continuous scalar function f that is 1 on $[a + \eta, \infty)$ and 0 on $(-\infty, a + \eta/2]$. Then

$$v(x) = f(B(x))v$$

depends continuously on x , equals v at x_0 , and belongs to E_x for nearby x . This proves lower semicontinuity. Lemma 2 supplies an orthonormal frame $Q(x)$ in E_x . Since every eigenvalue of $B(x)|_{E_x}$ is strictly larger than a , its compression to the range of $Q(x)$ is strictly larger than aI_n . $\square$

Remark 4. *No spectral projection is asserted to be continuous. The proof uses only lower semicontinuity of the high spectral field, which survives arbitrary eigenvalue crossings and rank jumps.*

3 Stable-rank count and factorization

Proof of Theorem 1. Since every column of $A(x)$ has norm at most $\|A(x)\|$, necessarily $0 < \theta \leq 1$. Put

$$B(x) = A(x)^* A(x), \quad a = \frac{\theta^2}{4}.$$

All eigenvalues of $B(x)$ lie in $[0, 1]$, while

$$\operatorname{tr} B(x) = \sum_{i=1}^N \|A(x)e_i\|^2 \geq N\theta^2.$$

Let $r(x)$ be the number of eigenvalues strictly larger than a . Bounding those $r(x)$ eigenvalues by 1 and the rest by a gives

$$N\theta^2 \leq \operatorname{tr} B(x) \leq r(x) + (N - r(x))a.$$

Therefore

$$r(x) \geq \left\lceil \frac{N(\theta^2 - a)}{1 - a} \right\rceil = \left\lceil \frac{3N\theta^2}{4 - \theta^2} \right\rceil = q.$$

Suppose first that $1 \leq n \leq q - d$. Lemma 3, applied to $X = \mathbb{R}^d$, gives a continuous orthonormal frame $R(x)$ such that

$$C(x) := R(x)^* A(x)^* A(x) R(x) > \frac{\theta^2}{4} I_n.$$

Define

$$L(x) = C(x)^{-1} R(x)^* A(x)^*.$$

Then L is continuous and $LAR = I_n$. Moreover $\|R(x)\| = 1$, and $L(x)$ is the canonical left inverse of $A(x)R(x)$. Hence

$$\|L(x)\| = \frac{1}{s_{\min}(A(x)R(x))} < \frac{2}{\theta}.$$

It remains only to cover the case in which $m = 1$ because $q - d \leq 0$. Take $R(x) = e_1$ and

$$L(x) = \frac{(A(x)e_1)^*}{\|A(x)e_1\|^2}.$$

Then $LAR = 1$, $\|R\| = 1$, and $\|L\| \leq 1/\theta$. This proves every assertion. $\square$

Corollary 5 (the source's diagonal formulation). *If the hypotheses instead say $|a_{ii}(x)| \geq \delta > 0$, the conclusion of Theorem 1 holds with $\theta = \delta$.*

4 Literature and novelty boundary

Question 1 of the source is no longer open. Fan et al. [3] prove the one-dimensional $\theta^2 N$ estimate and the resulting factorization in their Theorem 3.10. Their final section explicitly lists analogous results for a multidimensional domain as future work. The theorem above addresses that surviving question; it uses unstructured spectral frames, corresponding to their first method rather than their coordinate-structured second method.

The sole indexed citation to Fan et al. is Müller–Tomilov [4]. It studies continuous and smooth compressions of infinite-dimensional operator functions on $[0, 1]$; it says that arbitrary compact metric domains are plausible but restricts its proofs to the interval. It neither states nor implies the finite-dimensional bound above. Bounded arXiv searches for the exact title, “restricted invertibility” with “continuous matrix”, and continuous families/choices of subspaces returned Fan et al. alone through 2026-08-11.

The conservative status is therefore a candidate full affirmative answer, pending human verification of Lemma 3. No optimality in the additive d -loss or in the numerical constant is claimed.

References

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